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OVERVIEW OF USAGE OF OPEN GOVERNMENT DATA IN BUILDING MACHINE LEARNING AND AI SOLUTIONS: A SYSTEMATIC LITERATURE REVIEW

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ABSTRACT

This paper provides a systematic literature review on integrating Open Government Data (OGD) with machine learning (ML) and artificial intelligence (AI) solutions. OGD refers to publicly accessible data produced or collected by governmental entities, promoting transparency and citizen engagement. The paper explores the application of OGD in various social development domains, highlighting its potential impact. It also highlights the difficulties and challenges that researchers face. The findings suggest that leveraging OGD in ML and AI has the potential to drive innovation, improve decision-making, and foster inclusive development across sectors. The review emphasizes the importance of responsible practices in utilizing OGD for ML and AI purposes.

KEYWORDS

Open Government Data, Open Data, Artificial Intelligence, Machine Learning, Literature Review

1. INTRODUCTION

Open Government Data (OGD) constitutes both a philosophical underpinning and a progressively evolving set of policies. Its core mission is to advance transparency, accountability, and the generation of value by facilitating universal access to government data for all (*Open Government Data - OECD*, n.d.). The core goal of OGD is that the citizens have the right to access data and information generated by governments in order to create common value and trust in the government (Fleiner, 2018).

In the context of our swiftly advancing digital environment, the fusion of data-driven technologies with principles of government transparency has opened up a promising frontier for innovation. The contemporary world is witnessing an unprecedented creation of data, a substantial share of which is administered by governmental institutions worldwide. In the development of advanced technologies such as machine learning (ML) and artificial intelligence (AI), new and innovative solutions are being created that can contribute to the community. As OGD is accessible to everyone, using advanced technologies makes it possible to create solutions that are for the common good of the community (Gao & Janssen, 2020).

Harnessing large data with advanced tech can create smart solutions, improving global lives. Recognizing tech's potential for sustainable development and optimizing existing systems through reengineering is vital. This paper, through systematic literature review (SLR) results, gives insight into the proposed solutions by researchers based on OGD, ML and AI. The aim of this paper is to identify fields that have been the focus of research so far, such as health or economy, and to assess the solutions proposed. Lastly, it provides insights into less developed fields. In order to fulfill our research goal, we designed and conducted a systematic literature review using Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flowchart. We set research questions, as well as inclusion and exclusion criteria.

The rest of the paper is organized as follows. Section 2 provides an overview of the concept of Open Government Data, machine learning and artificial intelligence. Section 3 presents Methodology, and in Section 4 are presented results and discussion. Lastly, in Section 5, we gave Conclusion about this SLR.

2. OPEN GOVERNMENT DATA

OGD stands for Open Government Data, which represents information created using public funds and offered for use or distribution according to licenses that allow derivative work and commercial use (Janssen et al., 2012). Open government initiatives promote transparency and engagement to enable effective oversight, and in doing so, they also foster trust in the government from the citizens. The data must be available in formats that can be used without payment of fees or other restrictions, and at least one tool must be accessible for their processing (*Почетна - Отворени подаци*, n.d.). Data formats are diverse, with a prevalence of not proprietary and machine-readable file formats, such as CSV and JSON.

Open Data, especially OGD, offers diverse benefits, including improved public administration efficiency, private sector economic growth, and broader social welfare. It enhances performance, leading to more efficient public services and reduced unnecessary spending through cross-sector data sharing. Easy access to information and knowledge benefits the economy by fostering innovation and new business models. Society benefits from increased transparency and accessibility, promoting collaboration, participation, and social innovation (*What Is Open Data | Data.Europa.Eu*, n.d.) By signing the Open Government Directive at the White House in 2009 (*Open Government Directive*, 2009), the expansion and development of open data began.

It's important to emphasize that Open Data and Open Government Data are not the same concept. Data can be opened by non-governmental organizations and institutions that are not part of the government. OGD refers specifically to data that the government opens to all interested parties.

3. METHODOLOGY

The main focus of a systematic literature review (SLR) is to provide a comprehensive summary of the existing research and evidence on a specific research question or topic. For this SLR Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework was used. The aim of the PRISMA statement is to help authors improve the reporting of SLR, and PRISMA consists of a four-phase flow: 1. Identification, 2. Screening, 3. Eligibility and 4. Included (Moher et al., 2009). We used two databases for this literature review – Scopus and Web of Science. For searching papers, we used string: ("Open government data" or "OGD") AND ("machine learning" or "artificial intelligence").

Two research questions has been posed:

RQ1. What are the primary domains or sectors where Open Government Data is predominantly used for ML and AI solutions?

RQ2. What challenges and barriers do researchers face when integrating Open Government Data into ML and AI solutions?

The strategy for identifying primary studies for this literature review was established based on five inclusion and three exclusion criteria outlined below.

Inclusion criteria:

IC1. The main objective of the paper must be related to ML and AI solutions based on OGD primarily.

IC2. If there are additional data sources besides OGD, they must be in the form of Open Data.

IC3. The paper must describe ML/AI solutions over data.

IC4. The paper must be written in English.

IC5. Papers published between 2017 and 2023.

Exclusion criteria:

EC1. Papers that are not related to research questions.

EC2. Duplicate papers.

EC3. Exclude papers that discuss OGD but do not relate it to ML or AI.

Figure 1 presents PRISMA workflow for SLR. The primary data considered for the papers primarily included the title, abstract, keywords, year, and document type. Subsequently, the exclusion of papers was carried out through a thorough examination of the complete paper, within the scope of the third part of the PRISMA statement regarding eligibility.



Figure 1. PRISMA workflow

4. RESULTS AND DISSCUSION

After defining the search string, 105 papers were obtained from the Scopus database, and 308 papers from Web of Science. Initially, it was necessary to remove duplicates, which totaled 38. This left 375 papers for the next screening phase. Titles and abstracts of the papers were reviewed, resulting in the exclusion of 303 papers deemed unsuitable for further research. This left 72 papers for the third-part, eligibility assessment. Access was not granted for 6 papers, leading to their exclusion, while 41 papers were rejected based on the defined inclusion and exclusion criteria mentioned in Section 3. In the end, a total of 25 papers were included in the literature review, consisting of 15 articles, 9 conference papers, and one book chapter.

Based on the IC5, papers from the period of 2017 to 2023 were reviewed. One relevant paper from 2017 and 2018 was included, as well as four papers from 2021 and 2023 until August. Additionally, five papers from 2019, 2020, and 2022 each were included in the literature review.

Table 1. Review of the domains that the papers address

Domain	Doman definition	Papers
Economy	Finance	(Mashkova et al., 2017)
	Taxes	(Cha, 2020)
	Real estate	(Karamanou et al., 2022), (Tchuente & Nyawa, 2022)
	Business	(Hopf & Riechel, 2018)
Environment	Agriculture	(Colombo-Mendoza et al., 2022), (Amani et al., 2020)
	Human-elephant conflict	(Gunawansa et al., 2023)
	Temperature maps	(Zumwald et al., 2021)
	Wave height	(Ahmad & Mansurova, 2021)
Policy	Public procurements	(Niessen et al., 2020), (García Rodríguez et al., 2020)
	Public discussions	(Kim et al., 2019)
Traffic	Traffic injuries	(Boonserm & Wiwatwattana, 2021)
	Traffic jam	(Brimos et al., 2023)
Health	Air pollution	(Avvari et al., 2023), (Sethi & Thakur, 2021)
General	Missing persons	(Gomes et al., 2023)
	Urbanism	(Milojevic-Dupont et al., 2020)
	Simplifying OGD based on ML/AI	(Gazzaz & Rao, 2019), (Guberović et al., 2022), (Chen et al., 2019), (Metsker et al., 2019), (Ronzhin et al., 2019), (Stylianou et al., 2022)

Table 1 summarizes the domains with the highest number of papers. Governments globally share datasets on various portals, and selected domains cover specific subfields. For instance, under economics, solutions address finance, taxes, business, and real estate. Other domains are similarly divided, including general ones like missing persons and urbanism, along with solutions applicable to OGD, such as smart searches.

RQ1: It can be clearly concluded that the most attention is currently devoted to creating solutions in the field of environmental preservation and economics. It is concluded that structured data, as mandated by OGD, are effectively used for predictions and statistics, and all proposed solutions can contribute to the community on a daily basis. Though other areas are touched upon, there's a lack of studies on OGD-based smart solutions in education.

RQ2. The majority of authors emphasize that their main obstacles in developing such solutions are poorly structured data, complex, outdated, or unusable, despite being fundamental concepts of open data. (Gazzaz & Rao, 2019), (Mashkova et al., 2017) say that one of the biggest challenges for implementing a solution is the lack of shared data models, even when it comes to OGD at the level of a single state. Authors in (Gomes et al., 2023) highlight issues of data duplication and completeness, emphasizing the need for consistent updates by the government to foster value-added solutions (Brimos et al., 2023). Other authors didn't pinpoint data issues, but prep work for combining datasets is inevitable for effective algorithm application.

5. CONCLUSION

Through a SLR, we explored current solutions created using OGD worldwide. It is evident that increasing attention is being given to developing solutions using OGD for the benefit of communities, as reflected in the growing number of published papers each year, as mentioned in Section 4.

However, there are still domains, such as open education, where solutions need further development. Researchers also frequently encounter issues related to data obtained from official portals. There is a need for data standardization for publication, as well as ensuring the timeliness, accuracy, and reliability of the data. Raising awareness about the contributions of OGD to the community and promoting solutions for the common good based on advanced technologies is essential. With existing knowledge, ideas, and solutions, it is possible to enhance existing information systems with added value using ML and AI in context of OGD. The continuation of the research involves further analysis of concepts and the application of technologies on OGD, deeper examinations, and comprehensive insights into its multifaceted utilization.

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